

Name: _____

SOLUTIONS

Date: _____



Year 8 Mathematics

Investigation 1, 2015

Topic – Surface Area of Prisms

Validation Section

Total Time:

30 minutes

Weighting:

7.5% of the year

Equipment:

Take-Home Section

Calculators and rulers are allowed

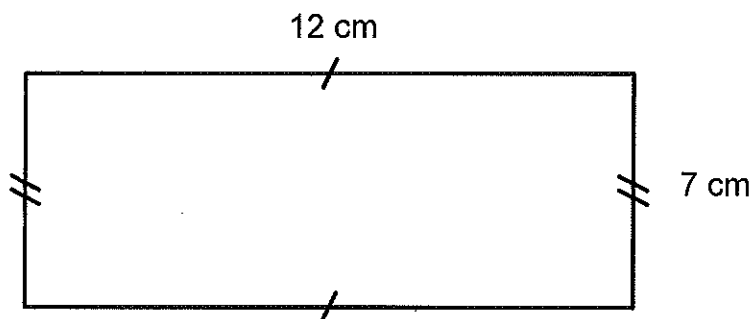
30 = _____ %
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Calculate the areas of these shapes with full working out.

***Only 2 of 5 marks will be awarded with working out ***

Question 1 (5 marks)

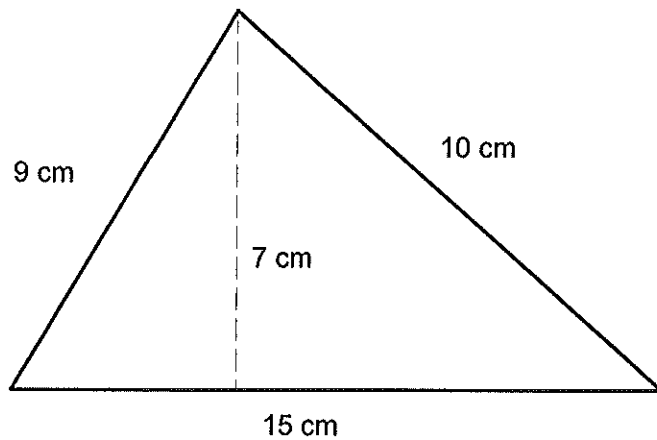
Calculate the area of this rectangle (show all working out)



$$\begin{aligned} A &= L \times W \checkmark \\ &= 12 \times 7 \checkmark\checkmark \\ &= 84 \text{ cm}^2 \checkmark\checkmark \end{aligned}$$

Question 2 (5 marks)

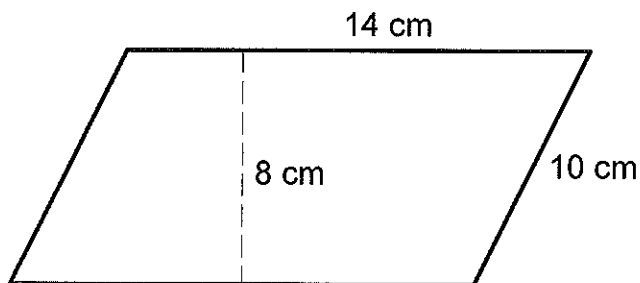
Calculate the area of this Triangle (show all working out)



$$\begin{aligned} A &= \frac{1}{2} \times PH \times B \checkmark \\ &= \frac{1}{2} \times 7 \times 15 \checkmark\checkmark \\ &= 52.5 \text{ cm}^2 \checkmark\checkmark \end{aligned}$$

Question 3 (5 marks)

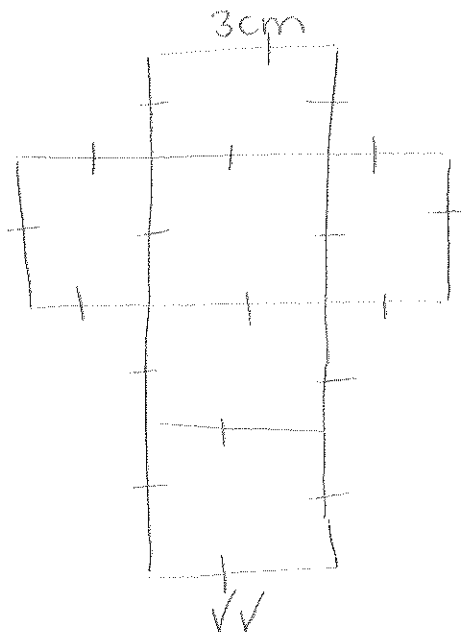
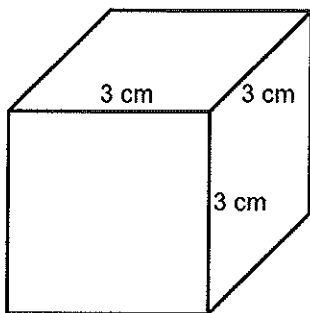
Calculate the area of this Parallelogram (show all working out)



$$\begin{aligned} A &= PH \times B \checkmark \\ &= 8 \times 14 \checkmark\checkmark \\ &= 112 \text{ cm}^2 \checkmark\checkmark \end{aligned}$$

Question 4 (5 marks)

Draw a net of this cube and calculate its total surface area

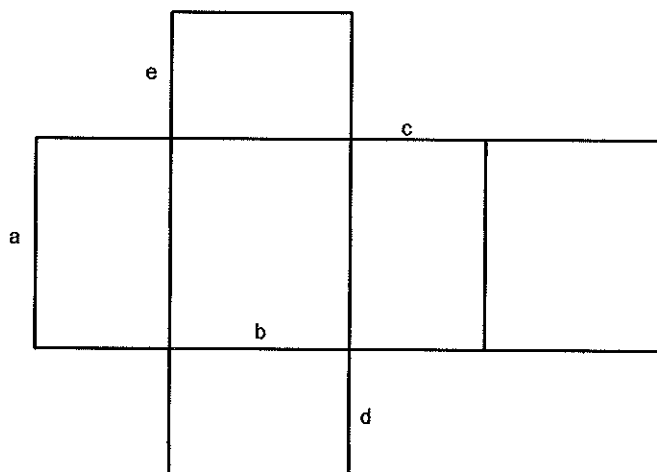
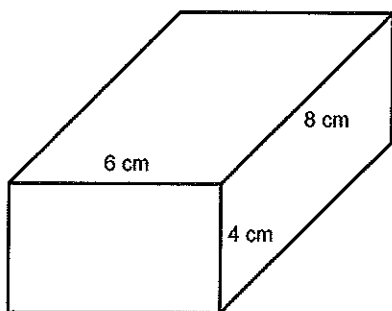


$$A = 3 \times 3 = 9 \text{ cm}^2 \checkmark$$

$$SA = 9 \times 6 = 54 \text{ cm}^2 \checkmark \checkmark$$

Question 5 (5 marks)

Below is an isometric drawing of a rectangular prism with its net next to it

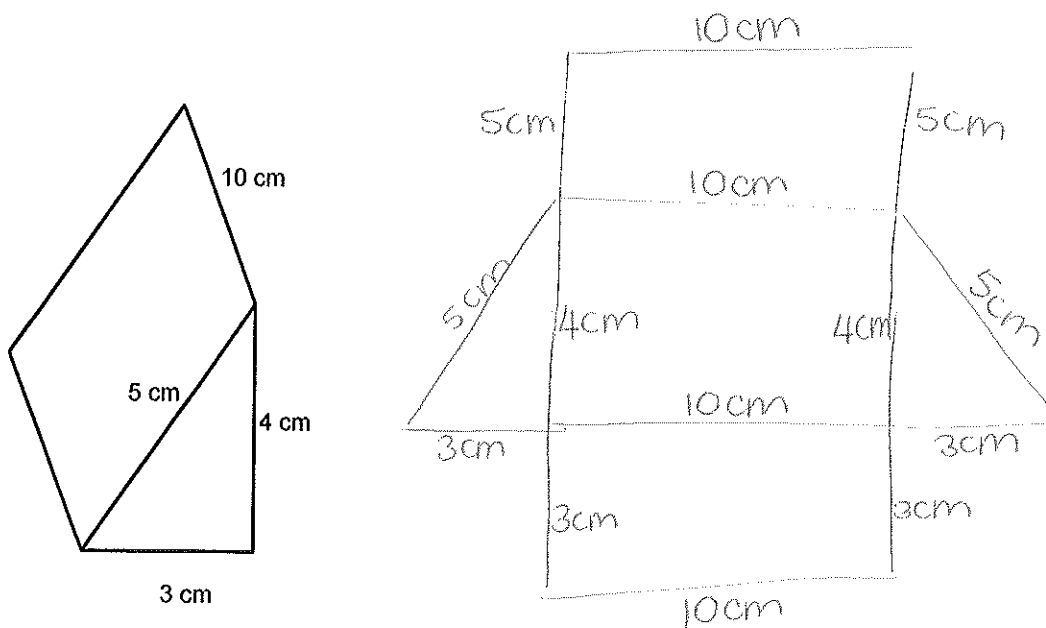


Write down the lengths of the labeled sides

a	8 cm	✓
b	6 cm	✓
c	4 cm	✓
d	4 cm	✓
e	4 cm	✓

Question 6 (5 marks)

Calculate the surface area of the triangular prism below (show all your working out)



$$\begin{aligned} SA &= 5 \times 10 + 2 \times \left(\frac{1}{2} \times 4 \times 3 \right) + 4 \times 10 + 3 \times 10 \quad \checkmark \checkmark \checkmark \\ &= 132 \text{ cm}^2 \quad \checkmark \checkmark \end{aligned}$$